

CPS I — Exercise Sheet 9

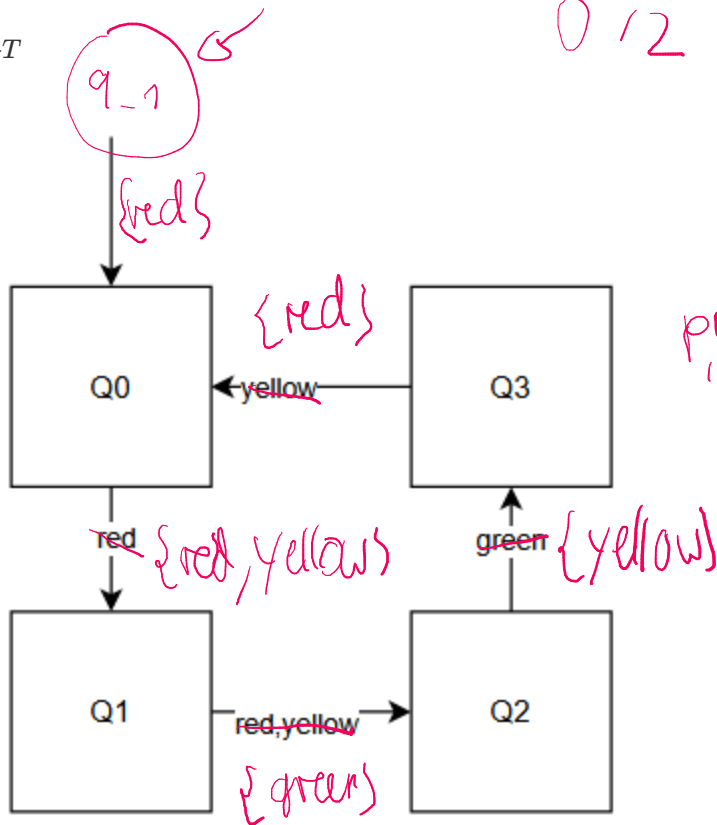
University of Freiburg · Winter Term 2025/26

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Exercise 1: Model Checking

1. A_T



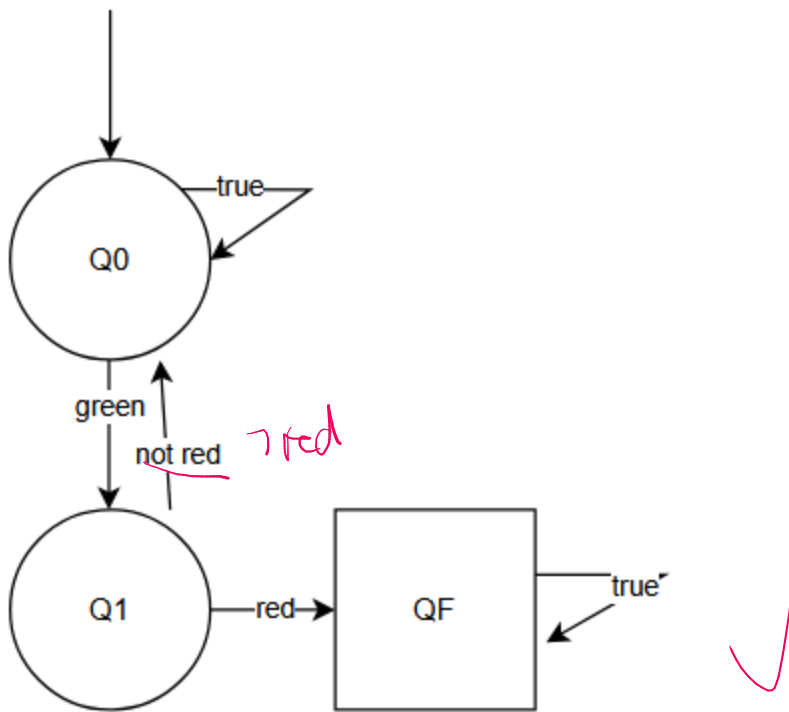
propositional formula
 "red" stands for the
 sets $\{red\}, \{red, yellow\}, \dots$
 same for "yellow", "green".

A_T does not accept
 exactly $\text{pref}(\text{Traces}(t))$

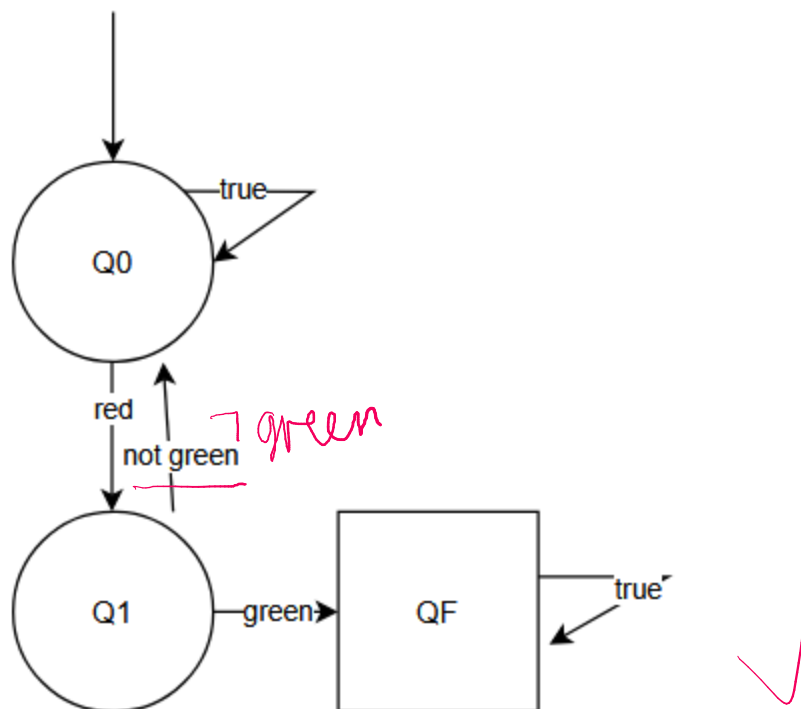
2. Bad Prefixes

A_{P1}

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A_{P2}

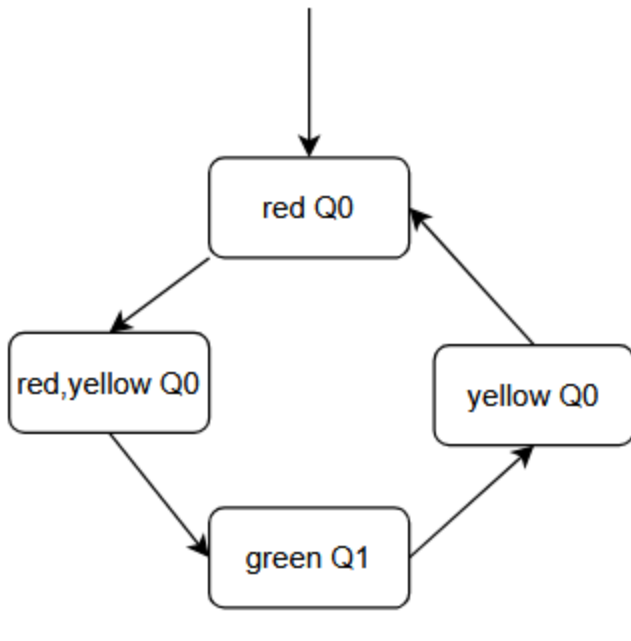


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3. Intersection

Intersection of A_T and A_{P1}
no accepting state can be reached

Look up again how
the intersection
automaton is constructed



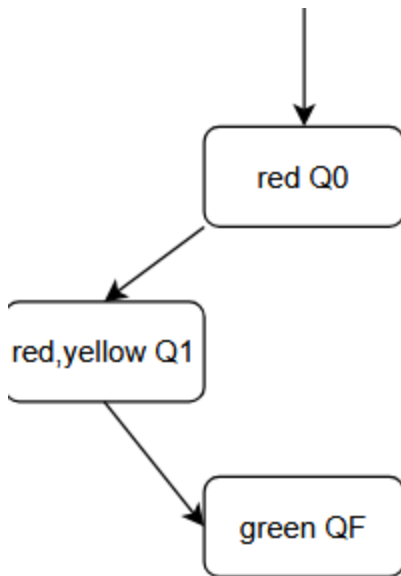
Intersection of A_T and A_{P2}

Is accepting state can be reached

$$\mathcal{L}(A_T \cap A_{P2}) \neq \emptyset$$

So TS doesn't satisfy P2

Counter Example: {red}{red,yellow}{green}



Exercise 2: Safety & Liveness

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• 1

◦ Safety $E = \{\sigma \in \Sigma^\omega \mid a \in A_0\}$

◦ Liveness $E = \{\sigma \in \Sigma^\omega \mid \exists i \geq 0, a \in A_i\}$

• 2

We state the domain of i ($\mathbb{N}$)
- 1 (✓)

$A_0 A_1 \dots$
if you don't define a trace as " $A_0 A_1 \dots$ " you can't refer to A_0 .
we write a dot
- 1 for notation

- Safety $E = \{\sigma \in \Sigma^\omega \mid \forall i \geq 0, a \notin A_i\}$ ✓
- Liveness $E = \{\sigma \in \Sigma^\omega \mid \exists^\infty i \geq 0, a \in A_i\}$ ✓
- 3
 - Safety $E = \{\sigma \in \Sigma^\omega \mid b \in A_0\}$ ✓
 - Liveness $E = \{\sigma \in \Sigma^\omega\}$ ✓
- 4
 - Safety: No, because every trace violates safety property has a bad prefix ✓
 - Liveness $E = \{\sigma \in \Sigma^\omega \mid \exists i \geq 0, a \in A_i\}$

0.5 / 8 ✓ 1

Exercise 3: LT Properties

- $P_1 = \{\sigma \in \Sigma^\omega \mid \forall i \geq 0, a \in A_i \vee b \in A_i\}$
 - Invariant
 - Safety
 - not Liveness
- $P_2 = \{\sigma \in \Sigma^\omega \mid \exists i \geq 0, (a \in A_i \wedge \forall j \neq i, a \notin A_j)\} \cup \{\sigma \in \Sigma^\omega \mid \forall i \geq 0, b \notin A_i\}$
 - not Invariant
 - not Safety
 - not Liveness
- $P_3 = \{\sigma \in \Sigma^\omega \mid \forall i \geq 0 (a \in A_i) \implies (b \notin A_{i+1})\}$
 - not Invariant
 - Safety
 - not Liveness

You can't refer to A_i / A_j if you don't define a trace as a sequence $A_0 A_1 \dots$ - 0.25

We write sets of traces like this:

$$\{A_0 A_1 \dots \in (\Sigma^\omega)^{\mathbb{N}} \mid \forall i \in \mathbb{N}, a \in A_i \vee b \in A_i\}$$

trace as sequence
domain of trace
domain of i
dot after quantifier

no explanations at all : 0.25 x 9

$P_4 - P_8 :$